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B.Tech. Degree III Semester Regular/Supplementary Examination November 2023

IT 19-204-0305 DATA STRUCTURES AND ALGORITHMS IN C
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Implement linear data structures such as arrays, linked lists, stack and queue with their related operations.
 CO2: Implement non linear data structures such as trees and graphs with their related operations.
 CO3: Implement various sorting and searching techniques.
 CO4: Identify suitable data structure and design technique for developing algorithm to solve a given problem.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A (Answer **ALL** questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a)	A stack is a linear data structure. Analyse the statement.	3	L1	1	1,2,3,4
(b)	When the array is sorted, binary search works better than linear search. Justify the statement based on time complexity.	3	L2	3	1,2,3,4
(c)	What is the worst case time complexity for inserting a new node in a double linked list? Analyse with an example list.	3	L4	1	1,3,12
(d)	Demonstrate the implementation of a priority queue using a linked list. What is the advantage of a priority queue over queue?	3	L3	4	1,5,12
(e)	Construct a max heap for the given array of elements- 2, 44, 62, 33, 7, 80, 67, 22, 56, 81, 98, 34	3	L3	2	1,5,12
(f)	What is the worst case time complexity for inserting a new node in (i) Binary tree (ii) Binary search tree (iii) AVL tree	3	L3	2	1,3,12
(g)	(i) All graphs are trees (ii) All trees are graphs (iii) Some graphs are trees Which of the above statements are true? Validate using ADT definitions of tree and graph.	3	L3	2	1,3,12
(h)	What is a dynamic data structure? Explain with examples.	3	L1	4	1,3,12

PART B

(4 × 12 = 48)

- II. Apply bubble sort and insertion sort on a reverse sorted array and analyse both the algorithms based on time complexity.
- OR**
- III. Perform time complexity analysis of linear search and binary search on an unsorted array of 'n' elements. Find the best case and worst case time complexity for both algorithms. Assume that selection sort is used to sort the array.

(P.T.O.)

		Marks	BL	CO	PO
IV.	Demonstrate the implementation of queue data structure using array and linked list. Analyse the implementation based on time complexity.	12	L3	1	1,3,12
OR					
V.	Write an algorithm for converting a given infix expression to a postfix expression using stacks. Analyse the space complexity of the algorithm. Assume array based stack implementation.	12	L4	4	1,3,12
VI.	Write algorithms for insertion and deletion operations on an AVL tree with n nodes. Analyse the algorithms for worst case and best case time complexity.	12	L3	2	1,3,12
OR					
VII.	Create AVL trees using the following stream of input (nodes inserted in this order only). (i) 22, 27, 31, 10, 5, 15, 29, 19, 16, 11, 3, 4, 8 (ii) 89, 56, 78, 26, 45, 35, 9, 11, 59, 34, 40, 39, 67, 6, 2	12	L3	2	1,3,12
VIII.	Explain different methods of graph representation using a complete graph having ten nodes as an example.	12	L2	4	1,3,12
OR					
IX.	Demonstrate different graph traversal algorithms with an example graph having ten vertices.	12	L2	4	1, 3, 12

Bloom's Taxonomy Levels

L1 - 5%, L2 - 22.5%, L3 - 50%, L4 - 22.5%.
